

SYNTHETIC DATASET

Core Features (Common across each datasets)

1. Dimensions

Dimensions are how we group and analyze data—by channel, platform, user, language, or time. They help us break down large data into meaningful segments so we can clearly see who is performing well, which platforms work best, and what content connects with the audience.

2. Uploaded Count

This shows how many videos are uploaded within a given period. It reflects consistency and output, and helps us understand whether growth is coming from posting more content or from creating better content.

3. Uploaded Duration

This represents the total length of all uploaded videos combined. It gives a sense of how much content is being produced and whether the effort put into creating long or multiple videos is actually paying off.

4. Views

Views indicate how many times content has been watched. They show how effective the content is at attracting attention, but don't guarantee that viewers stayed engaged throughout.

5. Likes, Comments, and Shares

These interactions reflect how audiences respond to content. Likes show approval, comments indicate deeper engagement, and shares highlight content that people find valuable enough to spread to others.

6. Avg. Retention

This measures how much of the video viewers watch on average. High retention suggests engaging and relevant content, while low retention signals that viewers are losing interest early.

7. Total Watch Time in Hours

This captures the total time viewers spend watching content. It is a key indicator of overall performance, showing how well the content holds attention and contributes to long-term audience engagement.

Mechanism ,Description , Distribution , Plot's for each dataset

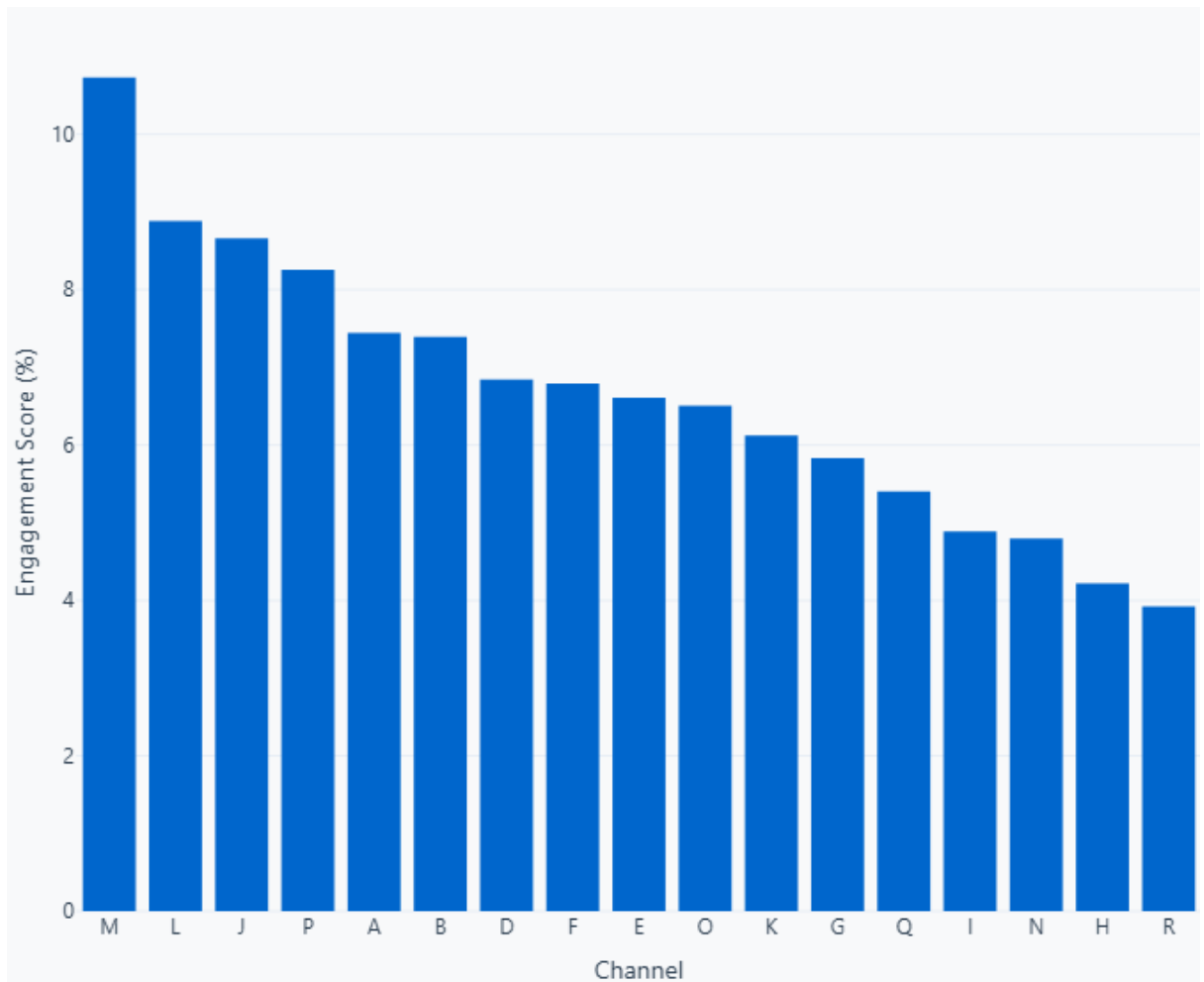
1 synthetic_channel_engagement

- **Mechanism:-** This system works like a smart hierarchy that builds channel-level performance from individual video data. It first distributes

reach (views and uploads) in a way where a few top performers take most of the attention. Then, engagement metrics like likes, comments, and shares are scaled proportionally to reflect realistic audience behavior. Finally, content quality is added using a balanced distribution for retention, ensuring that everything stays logically consistent with the total watch time.

- **Description:-** This gives a high-level overview of how each major channel or brand is performing. It helps compare different channels (like Channel A vs Channel B) to understand which ones are driving the most value and impact.
- **Features in this dataset:**
 - Channel: The main brand or network where content is published
 - Uploaded Count / Uploaded Duration: How much content each channel produces
 - Views / Likes / Comments / Shares: The reach and audience interaction for each channel
 - Avg. Retention (%) / Total Watch Time (hrs): The overall content quality and its effectiveness in keeping viewers engaged.
- **Assumptions:-** The dataset assumes that a small number of top channels generate most of the traffic (similar to the 80/20 rule). It also assumes that engagement (likes, comments, shares) is closely tied to views, meaning higher views naturally lead to higher engagement, with only slight variation.
- **Distribution:-**
 - Views & Engagement: Most content gets low to moderate numbers, while a few pieces perform extremely well, creating a highly uneven distribution
 - Avg. Retention: More balanced and consistent across videos, centered around an average of roughly 45–50%, and not directly dependent on views

○ **Plot :-**



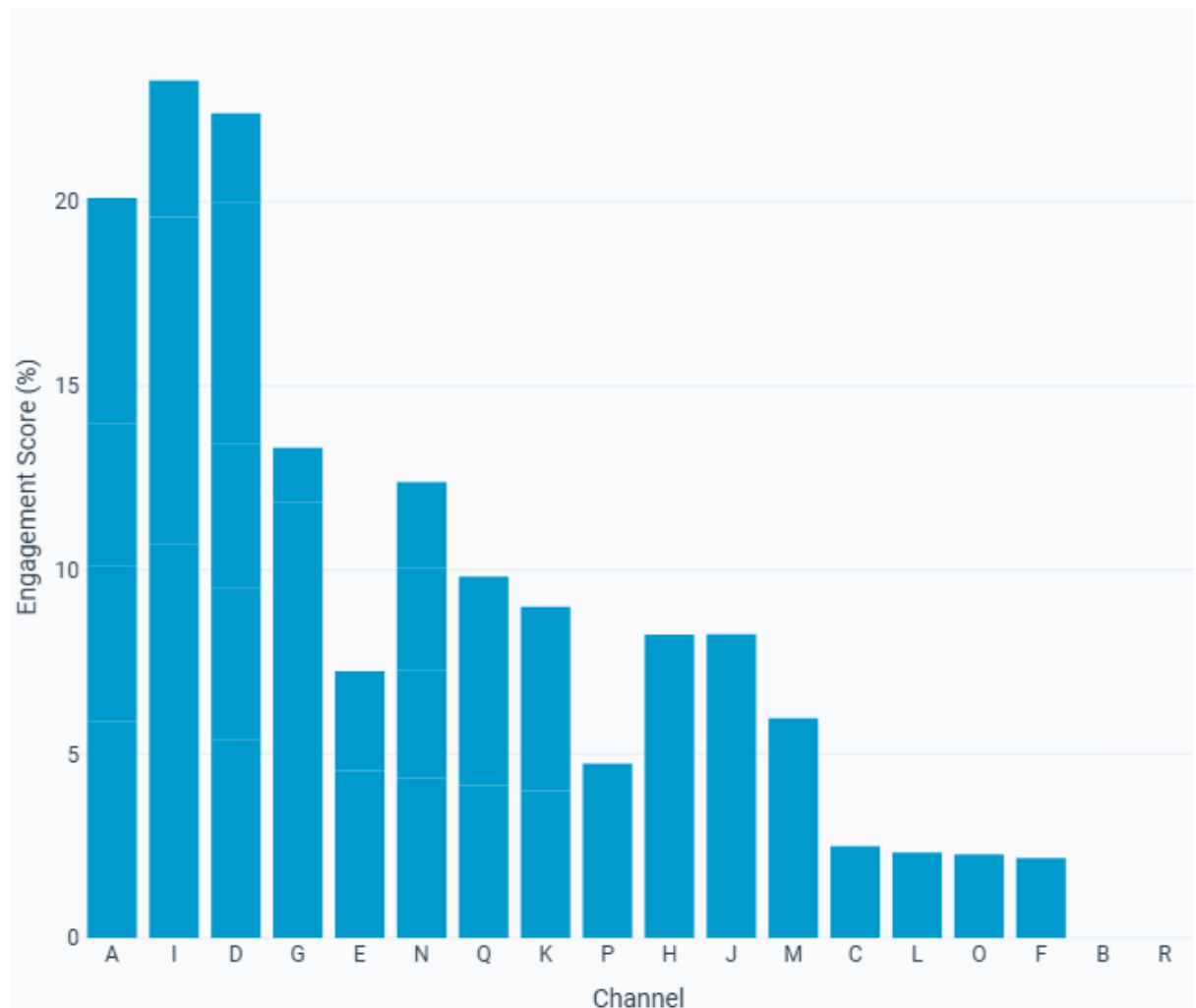
2 synthetic_channel_platform_engagement

- **Mechanism:-** This model simulates how content performs differently across platforms. It assumes that some platforms can make content go viral while others may not perform as well. Views are distributed in a way that reflects this unpredictability, and engagement (likes, comments, shares) is then scaled based on those views. Retention is added separately to represent how long people typically stay engaged, independent of how viral the content is. The focus here is purely on how reach turns into interaction, without considering content volume or total watch time.
- **Description :-** This dataset gives a deeper breakdown of performance across platforms. It shows how the same channel performs on different social media platforms like YouTube, Instagram, or LinkedIn. This helps in deciding where to focus efforts, invest more, or tailor content strategies.
- **Features :-**

- Channel: The main brand or network
- Platform: The social media platform (YouTube, Instagram, X, LinkedIn, etc.)
- Views / Likes / Comments / Shares: Platform-specific reach and engagement
- Retention (%): How well viewers on that platform stay engaged with the content

- **Assumptions:-** The model assumes that some platforms drive extremely high visibility while others may have minimal impact. Engagement is directly linked to views, meaning higher reach naturally leads to higher interactions. It also assumes that metrics like upload count and watch time are not relevant here, as the focus is only on how views convert into engagement.
- **Distribution :-**
 - Views & Engagement: Highly uneven—most content performs modestly, while a small portion achieves very high reach, especially on viral-friendly platforms
 - Retention: More stable and balanced, typically centered around the low-to-mid 40% range, and not strongly influenced by how viral the content is

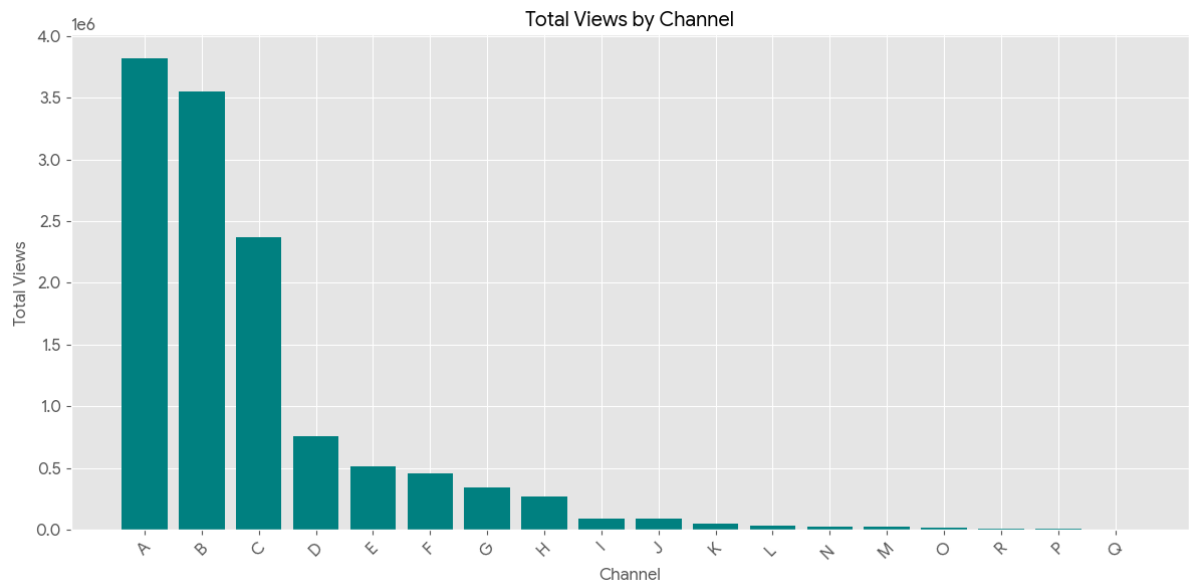
▪ **Plot:-**



3 synthetic_channel_user_engagement

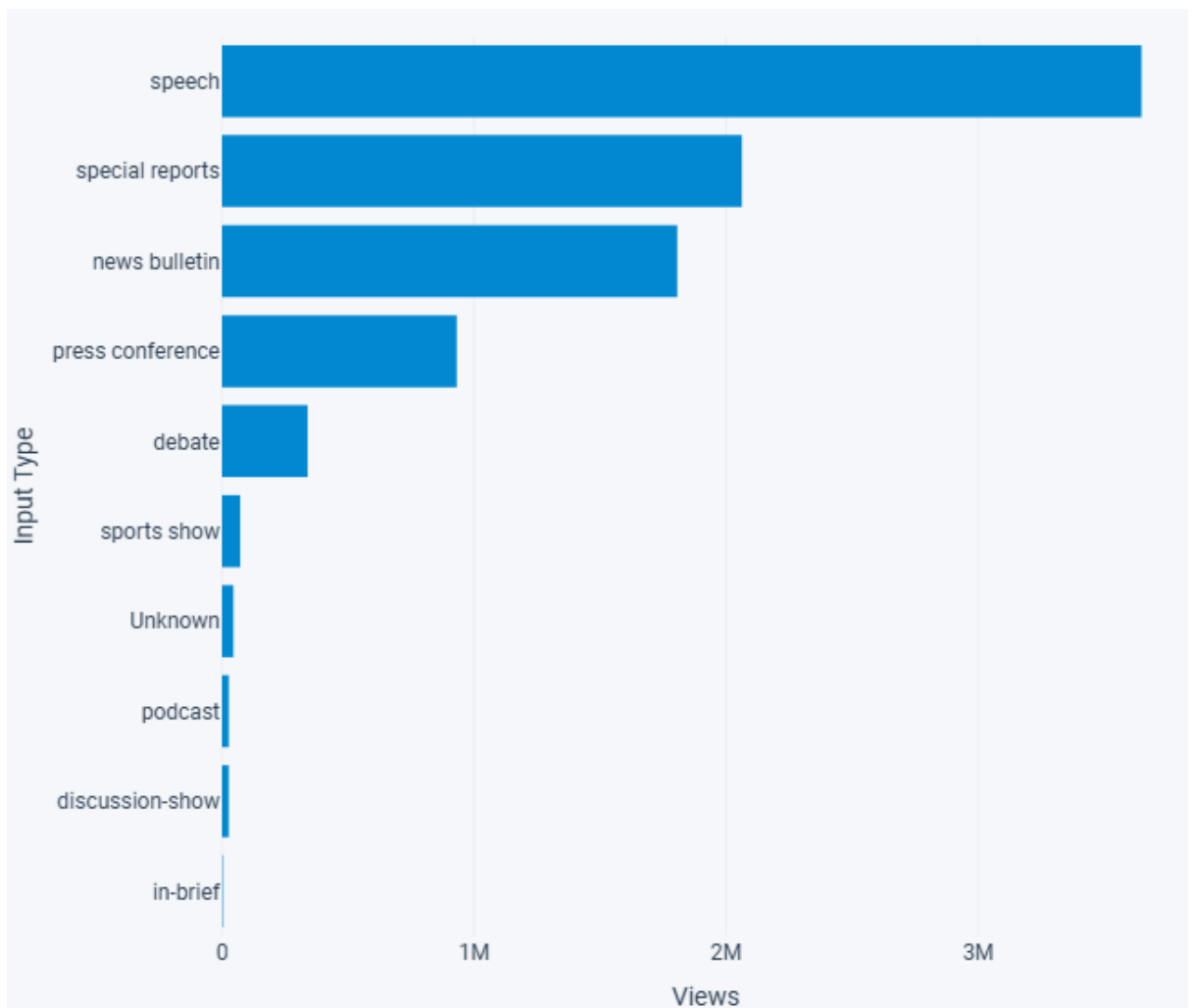
- **Mechanism:-** This model shows how content performs differently on each platform. Some platforms can make content go viral, while others may not. Based on this, views are distributed unevenly, and engagement (likes, comments, shares) grows along with those views. Retention is added separately to reflect how long people actually stay interested. The main focus here is understanding how reach turns into interaction, without worrying about how much content is uploaded.
- **Description :-** This dataset breaks down performance by platform. It shows how the same channel performs on different social media platforms like YouTube, Instagram, or LinkedIn. It helps in deciding where to focus more effort and which platform works best.
- **Features :-**
 - Channel: The brand or network
 - Platform: Where the content is posted (YouTube, Instagram, X, LinkedIn, etc.)
 - Views / Likes / Comments / Shares: Performance and engagement on each platform
 - Retention (%): How long viewers stay engaged on that platform.
- **Assumptions:-** It assumes that a few platforms drive most of the visibility, while others may not perform as strongly. Engagement is directly linked to views, so more views usually mean more likes, comments, and shares. It also ignores upload count and watch time to focus only on how views convert into engagement.
- **Distribution :-**
 - Views & Engagement: Very uneven—some content gets very high reach, while most stays low
 - Retention: More stable, usually around 40–45%, and not heavily affected by how viral the content is

- **Plot:-**



4 synthetic_input_type_engagement

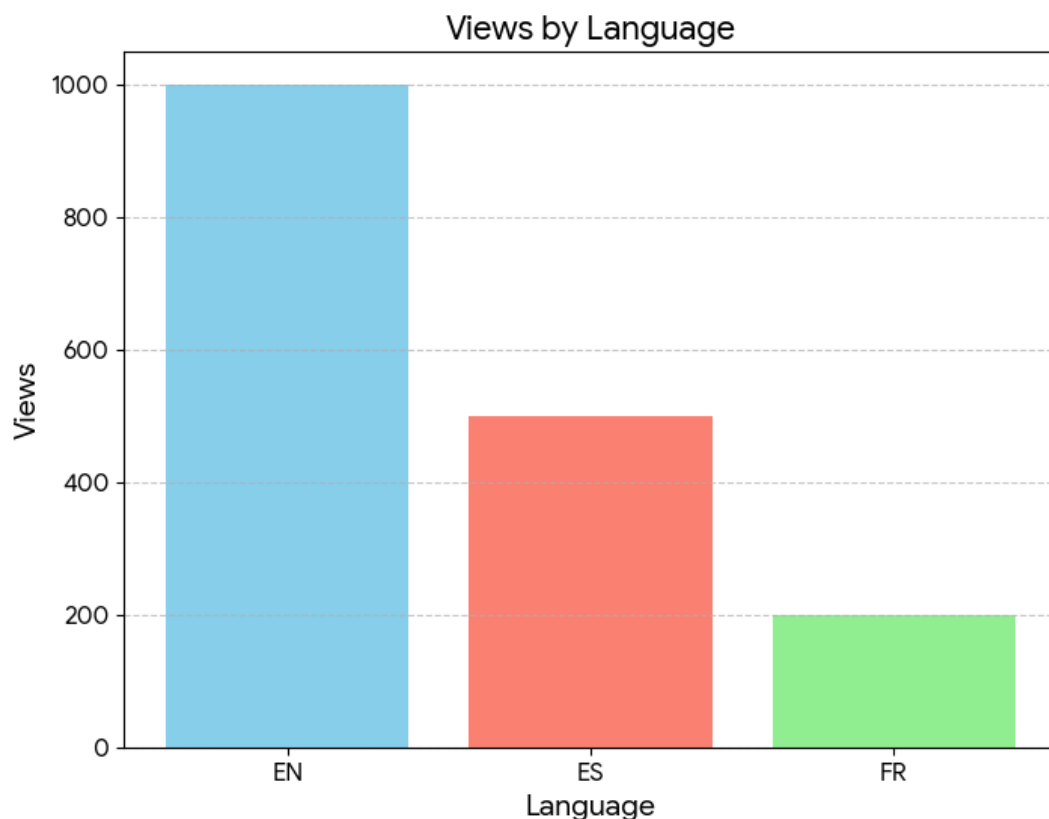
- **Mechanism:-** Think of this as showing how content behaves on different platforms. Some platforms can suddenly make a video blow up, while others give steady but lower reach. Based on that, views vary a lot, and engagement (likes, comments, shares) naturally follows those views. Retention is added separately to reflect how long people actually stay interested. The idea is to focus only on how people discover and interact with content.
- **Description:-** This dataset helps you compare how the same channel performs across different platforms. It clearly shows which platform is working best and where you should focus more effort.
- **Features :-**
 - Channel: The brand or content source
 - Platform: Where the content is posted (like YouTube, Instagram, etc.)
 - Views / Likes / Comments / Shares: How the content performs on each platform
 - Retention (%): How long viewers stay engaged.
- **Assumptions:-** It assumes that not all platforms perform equally—some drive most of the reach, while others don't. Engagement mostly depends on views, so more views usually mean more interaction. It ignores upload count and watch time to keep the focus on performance and engagement.
- **Distribution :-**
 - Views & Engagement: A few posts perform really well, while most stay average or low
 - Retention: Fairly consistent across platforms, usually around 40–45%
- **Plot:-**



5 synthetic_language_engagement

- **Mechanism:-** This model shows how content performs across different languages. It assumes that most of the views and uploads come from a few major languages like English and Hindi, while other languages get much less attention. Engagement (likes, comments, shares) grows along with views. Retention is handled a bit differently—it reflects that popular languages tend to have more stable viewing behavior, while smaller languages can be more unpredictable. Everything is then combined to keep the total watch time realistic.
- **Description:-** This dataset focuses on audience language. It helps understand how content performs in different languages and is useful for deciding whether to create regional content or invest in translations.
- **Features :-**
 - Language: The language of the content (like English, Hindi, Spanish, etc.)
 - Uploaded Count / Uploaded Duration: How much content is created in each language

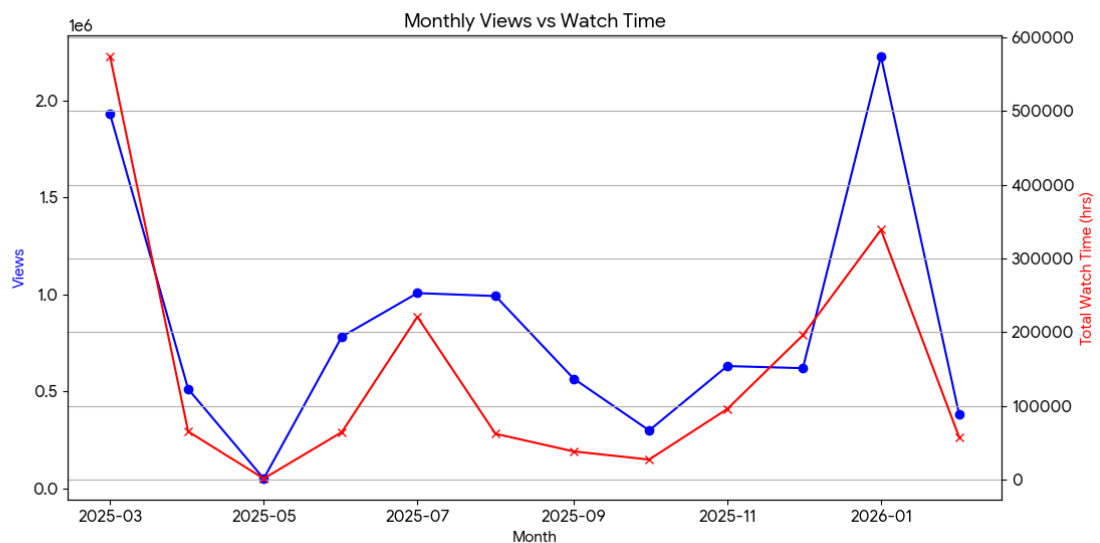
- Views / Likes / Comments / Shares: Audience size and engagement by language
- Avg. Retention (%) / Total Watch Time (hrs): How deeply audiences in each language consume the content
- **Assumptions:-** It assumes that most users come from one or two main languages, while other languages have much smaller audiences. Engagement is directly linked to views, so higher views lead to higher interaction.
- **Distribution :-**
 - Views & Engagement: Most of the traffic is concentrated in a few major languages, while the rest get very little
 - Retention: Slightly uneven—major languages tend to have steady retention (around 35–40%), while smaller languages can vary more
- **Plot:-**



6 synthetic_monthly_engagement

- **Mechanism:-** This model looks at how performance changes over time. It assumes that views don't just depend on how much content is uploaded—sometimes a single viral moment or seasonal trend can drive huge traffic. Engagement (likes, comments, shares) grows along with views, while retention stays fairly stable since audience behavior doesn't change drastically month to month. Everything is combined to keep the total watch time realistic and consistent over time.

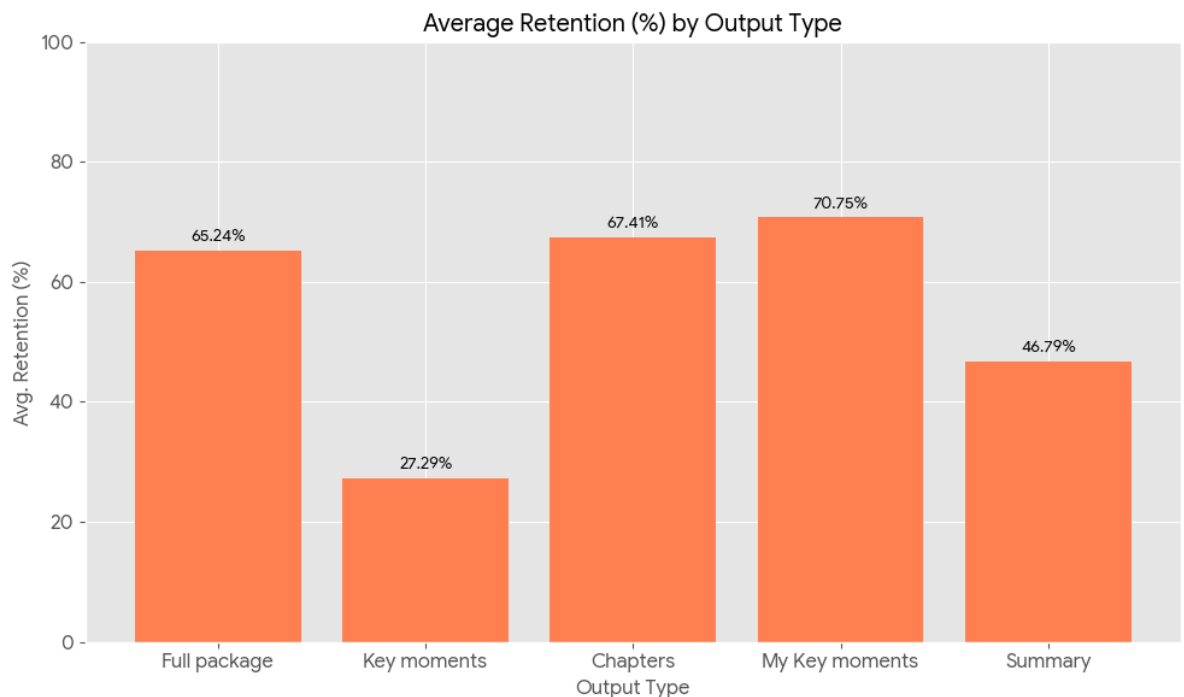
- **Description:-** This dataset focuses on performance across different months. It helps track growth, spot trends, and understand how specific campaigns or events impact results over time.
- **Features :-**
 - Month: The time period (e.g., Apr 2025)
 - Total Uploaded: Total content published in that month
 - Views / Likes / Comments / Shares: Overall reach and engagement for the month
 - Avg. Retention (%) / Total Watch Time (hrs): How well the audience stayed engaged during that period
- **Assumptions:-** It assumes that views are not strongly tied to how much content is uploaded. Instead, trends, seasons, or viral events play a bigger role. So even with fewer uploads, a month can perform really well if something goes viral.
- **Distribution :-**
 - **Views & Engagement:** More balanced compared to other datasets, with some spikes during peak months
 - **Retention:** Fairly stable, usually around 50%, showing consistent audience behavior over time
- **Plot:-**



7 synthetic_output_type_engagement

- **Mechanism:-** This model assumes that almost all the attention goes to the main video (the full package). Smaller features like summaries or chapters are used much less. Engagement (likes, comments, shares) mainly comes from the main video, while the smaller assets contribute very little. Retention can vary for these smaller pieces, but overall watch time is still dominated by the full video.

- **Description:-** This dataset looks at how users interact with different video formats—whether they watch the full video or just quick options like summaries or chapters. It helps understand actual viewing behavior and feature usage.
- **Features :-**
 - Output Type: Type of content (Full video, Summary, Chapters, etc.)
 - Uploaded Count / Uploaded Duration: Availability of each type
 - Views / Likes / Comments / Shares: Engagement for each format
 - Avg. Retention (%) / Total Watch Time (hrs): Time spent on full videos vs shorter formats.
- **Assumptions:-** It assumes that most users prefer watching the full video, while features like summaries and chapters are used very little.
- **Distribution :-**
 - **Views & Engagement:** Most of the activity is concentrated on the main video, with very little on smaller features
 - **Retention:** Fairly stable overall, usually around 50%, though it may vary slightly for shorter formats.
- **Plot:-**



8 synthetic_user_engagement

- **Mechanism:-** This model looks at performance at the individual creator level. It assumes that a small group of highly active creators produce most of the content and generate the majority of views. Engagement (likes, comments, shares) grows along with those views, while retention varies naturally based on content quality. All of this is combined to give a clear and realistic picture of each creator's overall impact.
- **Description:-** This is like a performance report for each creator. It shows how well an individual is doing across all channels and helps identify top

performers, measure productivity, and support decisions like rewards or recognition.

- **Features :-**

- User: The creator or team member
- Uploaded Count / Uploaded Duration: Total content created by them
- Views / Likes / Comments / Shares: Their overall reach and engagement
- Avg. Retention (%) / Total Watch Time (hrs): How engaging and effective their content is

- **Assumptions:-** It assumes that not all creators perform equally—some contribute much more than others. Engagement is directly linked to views, and watch time depends on both how many people watch and how long they stay.

- **Distribution :-**

- Views & Engagement: A few creators perform very well, while most stay in the middle range
- Retention: Fairly consistent across creators, usually around 45–50%

- **Plot:-**

